CantataS: a Community-aware agent-based citation Simulator

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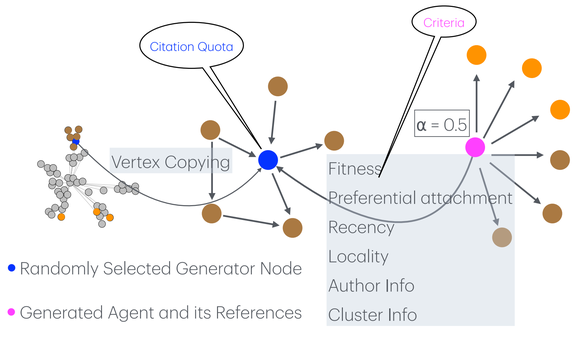
Motivation

- Understanding how, why, and what researchers cite is of broad interest
- Observations in the modern scientific enterprise are incommensurable with earlier studies
- Simulations under agent-based models complement social theory and mathematical models

Background

- Building upon prior work, we have developed a series of idealized agent-based models of citation [1, 2, 3, 4]
- These generative models return a citation network with nodes as articles and edges as citations
- CantataS adds community-awareness as a feature in addition to preferential attachment, fitness, recency, and author reputation

Methods



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- b1
- b2
- b3

Results

Limitations in prior work

- b1
- b2

Our goals

- Leverage invariant learning in causal effect estimation
- Capture **non-linear** patterns in the data using an autoencoder framework
- Use latent-variable modeling that can be robust even with the existence of confounding variables and **covariate shifts** within control units

Discussion

- We compared AE with state-of-the-art methods on CMAP dataset, which contains measurements of gene expression for more than 1 million samples.
- We measured accuracy in terms of root mean square error (RMSE), mean absolute error (MAE), and cosine similarity, which is an alignment measure.
- The training and evaluation are done in a leave-one-out manner. The boxplots show error (with mean and standard deviation) across 519 samples.

Observation: The current AE framework outperforms **5 out of 6** baselines in terms of cosine similarity, and **4 out of 6** baselines in terms of RMSE.

References

[1] G. Chacko et al. “An agent-based model of citation behavior”. In: *Applied Network Science* 11.1 (Jan. 2026).

[2] M. Park et al. “Very Large Scale Simulations of Network Growth with the Scalable Agent-Based Simulator for Citation Analysis with Sampling (SASCA-s)”. In: *Complex Networks & Their Applications XIV*. Springer Nature Switzerland, 2026, 3–14.

[3] M. Park et al. “Modeling the global citation network using the scalable agent-based simulator for citation analysis with recency-emphasized sampling (SASCA-ReS)”. In: *Scientometrics* 131.10 (Aug. 2026), 7209–7226.

[4] M. Park et al. *Modeling citations and cartels.* en. Accepted, Complex Networks & Their Applications XIV. 2026.